Local fields of characteristic p, limits of local fields of characteristic 0

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INTRODUCTION

D. Kazhdan introduced in the theory of representations of a reductive group over a local field the principle that the theory of a local field of characteristic $p > 0$ is the limit of the theories of local fields of characteristic 0 with the same residue field, when the absolute ramification index e tends to infinity.

According to the Langlands philosophy, one should expect the same phenomenon to appear for linear representations (of finite dimension) of the Galois group $\mathrm{Gal}(\bar{F}/F)$ of such a local field. In fact, on the Galois side, M. Krasner highlighted this phenomenon as early as the forties. See [3], [4], [5]. To study it, M. Krasner introduced the multiplicative quotient hyperfield of a valued field F by an ideal q of the ring $\mathcal{O}$ of the valuation. We assume $q \neq 0$ and $q \neq \mathcal{O}$. It consists of the set $F/(1+q)^*$ quotient of F by the equivalence relation $x \in (1+q) \cdot y$, equipped with the multiplication and (multivalued) addition deduced from those of F. For the description of an object (denoted $Q_q(F)$) which carries the same information and does not use a multivalued composition law, see the end of the introduction.

We will only consider complete valued fields, the valuation being real-valued (height 1). Let F be such a field, and F_i a sequence of such fields. We assume that F and the F_i are equipped with valuations having the same group $\Gamma \subset \mathbb{R}$ as set of values and, for $\gamma \in \Gamma$, we set $m_i^\gamma = \{x \in F_i \mid v(x) \geq \gamma\}$ and $m^\gamma = \{x \in F \mid v(x) \geq \gamma\}$. M. Krasner says that the F_i tend to F if, for a sequence $\gamma(i) \in \Gamma$ tending to $+\infty$, we are given isomorphisms from $F/(1+m^{\gamma(i)})^*$ with $F_i/(1+m_i^{\gamma(i)})^*$. For any ideal $q = m^\gamma$ of $\mathcal{O}_F$, if we set $q_i = m_i^\gamma$, we deduce for all i large enough an isomorphism from $F/(1+q)^*$ with $F_i/(1+q_i)^*$. If $P \in F[X]$ is a separable unitary polynomial, we know that the F-algebra $F[X]/(P)$ does not change (up to canonical isomorphism) when we perturb the coefficients of P sufficiently little. Newton's method indeed allows finding a unique zero of P near each root of the perturbed polynomial. If the F_i tend to F, we can use this fact to attach to a separable extension E of F an extension E_i of F_i for i large enough. M. Krasner shows that if E is Galois over F,

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E_i is so over F_i, with the same Galois group, and the structure of the ramification is preserved (for i large). Everything can already be read on $F/(1+\mathfrak{q})^*$, $\mathfrak{q}=\mathfrak{m}^\gamma$ for γ large, and $F/(1+\mathfrak{q})^*$ is isomorphic to $F_i/(1+\mathfrak{q}_i)^*$, for i large.

In the present article, we limit ourselves essentially to considering complete valued fields for a discrete valuation, with perfect residue field. If such a field is of characteristic $p>0$ and therefore isomorphic to a field of formal series $k((x))$, it is easy to approximate it by fields of characteristic 0. We specify the results of Mr. Krasner limited to the particular case considered. For each power $\mathfrak{q}=\mathfrak{m}^e$ $(e\geq 1)$ of the maximal ideal $\mathfrak{m}$ of the valuation ring $\mathcal{O}$ of F, we define a full subcategory $\mathbf{E}_e$ of the category $\mathbf{E}$ of finite separable extensions of F. It grows with e and $\mathbf{E}$ is the union of the $\mathbf{E}_e$. We show that $\mathbf{E}_e$ depends only on $F/(1+\mathfrak{m}^e)^*$. More precisely, we construct a category $\mathbf{E}(F/(1+\mathfrak{m}^e)^*)$ depending only on $F/(1+\mathfrak{m}^e)$ and a canonical equivalence of categories $\mathbf{E}_e\to\mathbf{E}(F/(1+\mathfrak{m}^e)^*)$. The full subcategory $\mathbf{E}_e$ of $\mathbf{E}$ has the following stability properties: with an extension E of F, it contains its normal closure and its sub-extensions; with two extensions E' and E'' of F, it contains their composite extensions. If $\bar{F}$ is a separable closure of F, such a subcategory $\mathbf{E}'$ of $\mathbf{E}$ corresponds to a subfield F' of $\bar{F}$ Galois over F, $\mathbf{E}'$ and F' determining each other mutually by the rules: F' is the union of the $E\subset\bar{F}$ in $\mathbf{E}'$, the finite sub-extension E of $\bar{F}$ is in $\mathbf{E}'$ if and only if $E\subset F'$. For $E'=\mathbf{E}_e$, the subfield F' of $\bar{F}$ is the subfield fixed by $\mathrm{Gal}(\bar{F}/F)^e$: the e-th ramification group in upper numbering ([6] IV §3 Remark 1 page 83). In other words, for a finite Galois extension E of F to be in $\mathbf{E}_e$, it is necessary and sufficient that the e-th ramification group in upper numbering $\mathrm{Gal}(E/F)^e$ be trivial. The category $\mathbf{E}_e$ determines the profinite group

$$Q^e\mathrm{Gal}(\bar{F}/F):=\mathrm{Gal}(\bar{F}/F)/\mathrm{Gal}(\bar{F}/F)^e$$

up to isomorphism, and even up to an inner automorphism (3.5).

The profinite group $Q^e\mathrm{Gal}(\bar{F}/F)$ depends therefore, up to inner automorphism, only on $F/(1+\mathfrak{m}^e)^*$. That such might be the case was suggested to us by class field theory. For F with finite residue field, this fur-

...indeed provides a canonical isomorphism of the abelianization of $Q^e\mathrm{Gal}(\bar{F}/F)$ with the profinite completion of $F^*/(1+\mathfrak{m}^e)^*$.

Among the compatibilities that we prove, let us point out the following:

a. Let F_i $(i=1,2)$ be as above. We assign an index i to objects relative to F_i. Let an isomorphism be given

$$F_1/(1+\mathfrak{m}_1^e)^*\simeq F_2/(1+\mathfrak{m}_2^e)^* \quad .$$

It provides an equivalence of categories

$$E_e(F_1)\simeq E(F_1/(1+\mathfrak{m}_1^e)^*)\simeq E(F_2/(1+\mathfrak{m}_2^e)^*)\simeq E_e(F_2) \quad .$$

If the extension E_1 of F_1 corresponds to the extension E_2 of F_2 (E_i in $E_e(F_i)$), the Herbrand functions for E_1/F_1 and E_2/F_2 are the same. Let us set $r=\psi_{E_1/F_1}(e)$. We construct an isomorphism $E_1/(1+\mathfrak{m}_1^r)^*\simeq E_2/(1+\mathfrak{m}_2^r)^*$ ($\mathfrak{m}_i$ being here the maximal ideal of the valuation ring of E_i) and establish various natural compatibilities for it.

b. If the F_i have finite residue fields, we prove the commutativity of the diagram

$$\begin{array}{ccc}
[Q^e\mathrm{Gal}(\bar{F}_1/F_1)]^{\mathrm{ab}} & \simeq & [Q^e\mathrm{Gal}(\bar{F}_2/F_2)]^{\mathrm{ab}} \\
\downarrow & & \downarrow \\
[F_1^*/(1+\mathfrak{m}_1^e)^*]^\wedge & \simeq & [F_2^*/(1+\mathfrak{m}_2^e)^*]^\wedge
\end{array}$$

In this diagram, the isomorphism in the first line is the quotient of the isomorphism (well-defined up to inner isomorphism) $Q^e\mathrm{Gal}(\bar{F}_1/F_1)\simeq Q^e\mathrm{Gal}(\bar{F}_2/F_2)$ deduced from the equivalence $E_e(F_1)\simeq E_e(F_2)$. The isomorphism in the second line is deduced from the isomorphism $F_1/(1+\mathfrak{m}_1^e)^*\simeq F_2/(1+\mathfrak{m}_2^e)^*$ and the vertical isomorphisms are those of class field theory for the F_i.

For fields F_i with algebraically closed residue fields, we have an analogous compatibility with geometric class field theory (3.8.1).

c. The isomorphism (well-defined up to inner isomorphism) $Q^e\mathrm{Gal}(\bar{F}_1/F_1)\simeq Q^e\mathrm{Gal}(\bar{F}_2/F_2)$ induces a bijection between the isomorphism classes of complex representations (continuous and of finite dimension) of the $\mathrm{Gal}(\bar{F}_i/F_i)$, trivial on $\mathrm{Gal}(\bar{F}_i/F_i)^e$. If the F_i have residue fields

finite, representations V_i whose isomorphism classes correspond give rise to the same factors L and ϵ (3.7).

Rather than Krasner's hyperfield $F/(1+q)^*$, whose multivalued addition makes us uneasy, we prefer to use the following object $Q_q(F)$, which carries exactly the same information: the graded ring, with degree group the group $\Gamma = F^*/\mathcal{O}^*$ of the valuation of F and with homogeneous components the

$$Q_q(F)^\gamma := \{x \in F \mid v(x) \geq \gamma\}/q\{x \in F \mid v(x) \geq \gamma\} ,$$

equipped with morphisms $\epsilon_{\delta,\gamma} : Q_q(F)^\gamma \to Q_q(F)^\delta$ for $\gamma \geq \delta$ deduced by passing to the quotient from the natural inclusions. Note that the multiplicative group of invertible homogeneous elements of $Q_q(F)$ is canonically isomorphic to $F^*/(1+q)^*$.

Suppose the valuation is discrete and normalized so that its group of values is $\mathbb{Z}$. The ring $Q_q(F)$ is then $\mathbb{Z}$-graded. Changing notation, we will write $Q_e(F)$ for what above would have been $Q_{\mathfrak{m}^e}(F)$ ($\mathfrak{m}$ the maximal ideal of the ring $\mathcal{O}$ of the valuation, $e \geq 1$). Let us set, for $n \in \mathbb{Z}$, $\mathfrak{m}^n := \{x \in F \mid v(x) \geq n\}$. We have

$$Q_e(F) = \bigoplus_{n\in\mathbb{Z}} \mathfrak{m}^n/\mathfrak{m}^{n+e} ,$$

the morphisms $\epsilon_{n',n} : \mathfrak{m}^n/\mathfrak{m}^{n+e} \to \mathfrak{m}^{n'}/\mathfrak{m}^{n'+e}$ for $n \geq n'$ being deduced from the inclusions $\mathfrak{m}^n \hookrightarrow \mathfrak{m}^{n'}$. The data of $Q_e(F)$ is equivalent to that of the ring $Q^0 := Q_e(F)^0 = \mathcal{O}/\mathfrak{m}^e$, of the free Q^0-module of rank one $Q^1 := Q_e(F)^1 = \mathfrak{m}/\mathfrak{m}^{e+1}$ and of the morphism of Q^0-modules $\epsilon = \epsilon_{0,1} : Q^1 \to Q^0$ from Q^1 onto the maximal ideal of Q^0. We reconstitute $Q_q(F)^n$ as being the n-th tensor power of the invertible Q^0-module Q^1 and the $\epsilon_{n',n}$ are deduced from ϵ. In the text, we will work with the triple (Q^0, Q^1, ϵ) rather than with $Q_{\mathfrak{m}^e}(F)$. We will denote this triple $Tr_e(F)$. The isomorphism class of Q^0 determines that of (Q^0, Q^1, ϵ), i.e. that of $Q_e(F)$, but the triple (Q^0, Q^1, ϵ) is not determined up to unique isomorphism by Q^0; it admits non-trivial automorphisms, but trivial on Q^0.

I had first thought to use, rather than $Q_e(F)$ or (Q^0, Q^1, ϵ), simply the quotient $Q^0 = \mathcal{O}/\mathfrak{m}^e$. The following example shows that this approach would provide less precise results. Let us take F with alge-

...ically closed and consider the category $\mathcal{E}_{\text{mod}}$ of tamely ramified extensions of F. One cannot reconstruct it up to unique equivalence up to unique isomorphism merely in terms of the residue field k of F. For $F = k((x))$, the automorphism $x \longmapsto \lambda x$ of F ($\lambda \in k$), trivial on k, would otherwise have a natural extension to $k((x^{1/N}))$ ($N > 1$ prime to the characteristic), which is absurd: for λ an N-th root of 1, this automorphism of order n cannot be extended into an automorphism of the same order. On the other hand, this category $\mathcal{E}_{\text{mod}}$ is entirely determined by $Q_1(F)$ or, what amounts to the same thing, by k, equipped with the k-vector space of rank one $\mathfrak{m}/\mathfrak{m}^2$: the functor which associates to a tame extension E of F, of degree d, the graded ring $Q_1(E) = \bigoplus \mathfrak{m}_E^n/\mathfrak{m}_E^{n+1}$, equipped with its natural structure of $Q_1(F)$-algebra for which $Q_1(F) \xrightarrow{\sim} \bigoplus_{d|n} \mathfrak{m}_E^n/\mathfrak{m}_E^{n+1}$, is fully faithful and one easily determines its essential image. A fortiori, the category $\mathcal{E}_{\text{mod}}$ is determined by $\mathcal{O}/\mathfrak{m}^2$, but working with $\mathcal{O}/\mathfrak{m}^2$ rather than with $(\mathcal{O}/\mathfrak{m}, \mathfrak{m}/\mathfrak{m}^2)$ would be less precise.

At the source of our usage of $Q_e(F)$ lies the following fact. A totally ramified extension E of F is obtained by adjoining to F a root of an Eisenstein equation $X^n + \sum a_i X^i = 0$, and it is more precise to consider the image $\bar{a}_i$ of a_i in $\mathfrak{m}/\mathfrak{m}^{e+1}$ than in $\mathcal{O}/\mathfrak{m}^e$. If one wished to treat complete valued fields with imperfect residue field, for which one can no longer factor every extension into an unramified extension, defined by a separable extension of the residue field, followed by a totally ramified extension, it is not clear to me that $Q_e(F)$ would remain preferable to $\mathcal{O}/\mathfrak{m}^e$.

1. PRESENTATION OF THE CHARACTERS

1.1.

Let us call a **truncated valuation ring** a local ring R whose maximal ideal $\mathfrak{m}$ is generated by one element, and is nilpotent. Let us define the **length** $\lg(R)$ of R as being that of the R-module R. It is also the smallest integer ℓ such that $\mathfrak{m}^\ell = 0$. We define the **valuation** of $x \in R$ by $v(x) := \sup\{i \mid x \in \mathfrak{m}^i\}$. It takes values in the interval $[0, \lg(R) - 1]$ of $\mathbb{Z}$, augmented by $\{\infty\}$ (with $v(0) = \infty$).

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Any truncated valuation ring R is the quotient of a discrete valuation ring V, which can be assumed to be complete, by a power of the maximal ideal. According to Cohen's theorems on the structure of

complete local rings, there indeed exists a regular complete local ring R_1 of which R is a quotient (EGA IV 1, 19.8.8). One can choose R_1 having the same Zariski tangent space as R (passing to a regular quotient). If one excludes the trivial case where R is a field, R_1 is then the desired ring V.

We will have to use triples (R, M, ε), formed of a truncated valuation ring with perfect residue field R, a free R-module of rank one M and an epimorphism ε of M onto the maximal ideal $\mathfrak{m}$ of R. We deduce from ε, for $r, s \in \mathbb{Z}$, $r < s$, a morphism

$$\varepsilon_{r,s} : M^{\otimes s} \longrightarrow M^{\otimes r} : x^{\otimes s} \longmapsto \varepsilon(x)^{s-r} \cdot x^{\otimes r}.$$

The isomorphism class of $S = (R, M, \varepsilon)$ depends only on that of R. More precisely, if $S' = (R', M', \varepsilon')$ is another triple, every isomorphism $u : R \xrightarrow{\sim} R'$ lifts to $\tilde{u} : S \xrightarrow{\sim} S'$. If π is a generator of M, lifting u to $\tilde{u}$ amounts to lifting $u\varepsilon(\pi) =$ a generator of the maximal ideal $\mathfrak{m}'$ of R' - to a generator π' of M'. If $\lg(R) > 1$, ε induces an isomorphism $M/\mathfrak{m}^{\lg(R)-1}M \xrightarrow{\sim} \mathfrak{m}$, similarly for S', and by Nakayama every lifting of $u\varepsilon(\pi)$ is a generator of M'.

To a triple $S = (R, M, \varepsilon)$, we associate a) the graded R-algebra $Q(S) := \bigoplus_{n\in\mathbb{Z}} M^{\otimes n}$. The choice of a generator π of M identifies it with the algebra of Laurent polynomials $R[T^{-1}, T]$. The $\varepsilon_{0,n} (n \geq 0)$ form a morphism from the subalgebra $\bigoplus_{n>0} M^{\otimes n}$ into R. b) The multiplicative group S^* of invertible homogeneous elements of $Q(S)$. The choice of a generator π of M identifies it with $\mathbb{Z} \times R^*$, by the map

$$\mathbb{Z} \times R^* \to S^* : (n, x) \longmapsto x \cdot \pi^{\otimes n}.$$

Let π be a generator of M. We define the <u>valuation</u> of $x = y\pi^{\otimes s} \in M^{\otimes s}$ by $v(x) = v(y)+s$. If $\varepsilon_{r,s}(x) \neq 0$, we have $v(x) = v\varepsilon_{r,s}(x)$. The valuation $v : M^{\otimes s} \longrightarrow \mathbb{Z}$ takes values in $[s, \lg R + s - 1] \cup \{\infty\}$.

1.2

We will call <u>local field</u> a complete valued field for a discrete valuation. The residue field will always be assumed to be perfect. For F a local field, we will denote by $\mathcal{O}_F$ its valuation ring, $\mathfrak{m}_F$ the maximal ideal of $\mathcal{O}_F$ and v_F the valuation, normalized to have $\mathbb{Z}$ as value group. If this creates no ambiguity, the index F will be omitted.

Let F be a local field and e an integer ≥ 1. We will denote by $\mathrm{Tr}_e(F)$ the following triple (R, M, ϵ) : $R = O/\mathfrak{m}^e$, $M = \mathfrak{m}/\mathfrak{m}^{e+1}$ and $\epsilon =$ natural projection of $\mathfrak{m}/\mathfrak{m}^{e+1}$ onto the maximal ideal $\mathfrak{m}/\mathfrak{m}^e$ of R. We have $\lg(R) = e$. Let π be a generator of $\mathfrak{m}$. For $s \in \mathbb{Z}$, we will denote by $\mathfrak{m}^s$ the fractional ideal $\hat{O} \cdot \pi^s \subset F$. It does not depend on the choice of π. The isomorphisms, also independent of π : $\mathfrak{m}^{\otimes s} \longrightarrow \mathfrak{m}^s : x\pi^{\otimes s} \longmapsto x\pi^s$ induce isomorphisms

$$\bigoplus M^{\otimes s} \xrightarrow{\sim} \bigoplus \mathfrak{m}^s/\mathfrak{m}^{s+e} \quad \text{and}$$

$$\mathrm{Tr}_e(F)^* \xrightarrow{\sim} F^*/(\text{units} \equiv 1 \pmod{\mathfrak{m}^e})^* \quad .$$

Via the isomorphisms $M^{\otimes s} \simeq \mathfrak{m}^s/\mathfrak{m}^{e+s}$, the $\epsilon_{r,s}$ are identified with the maps $\mathfrak{m}^s/\mathfrak{m}^{e+s} \longrightarrow \mathfrak{m}^r/\mathfrak{m}^{e+r}$ deduced from the inclusions $\mathfrak{m}^s \subset \mathfrak{m}^r$, and v on $M^{\otimes s}$ is deduced from the restriction to $\mathfrak{m}^s$ of the valuation of F.

The triples $\mathrm{Tr}_e(F)$ attached to F are, in substance, the "valued hyperfields quotients of F by a multiplicative divisor" that Krasner considers [5].

Every triple (R, M, ϵ) of 1.1 is isomorphic to one of those obtained by the procedure 1.2.

1.3 Let F be a local field with perfect residue field and f real ≥ 0. We will denote by $(\text{ext } F)^f$ the category of finite separable extensions E of F satisfying the following condition C_F^f. C_F^f. The normal closure E_1 of E over F satisfies $\mathrm{Gal}(E_1/F)^f = \{1\}$. For a list of conditions equivalent to C_F^f, see the appendix (A.6.1 and A.6.2).

The essential part of the article consists in (a) for $S = (R, M, \epsilon)$ as in 1.1, with $\lg(R) = e$, constructing a category $(\text{ext } S)^e$; (b) for F as above and e an integer ≥ 1, constructing an equivalence of categories

$$T : (\text{ext } F)^e \longrightarrow (\text{ext } \mathrm{Tr}_e F)^e \quad .$$

If F' and F'' are two local fields with perfect residue field, equipped with uniformizers π', π'', an isomorphism $u : O'/\mathfrak{m}'^e \longrightarrow O''/\mathfrak{m}''^e$ sending the class of π' onto that of π'' induces an isomorphism of $\mathrm{Tr}_e(F')$ with $\mathrm{Tr}_e(F'')$. From the equivalences T and from this isomorphism, one deduces an equivalence

$$(\text{ext } F')^e \longrightarrow (\text{ext } F'')^e \quad .$$

Here is a partial description of it. To an extension E' of F', satisfying $\mathcal{C}^e_{F'}$, and assumed for simplicity to be totally ramified, we will associate the following extension E'' of F''. One chooses a uniformizer x of E'. It satisfies an Eisenstein equation $P(x) = 0$. We write $P(X)$ in the form

$$P(X) = X^n + \pi' \sum a'_i X^i \quad .$$

The isomorphism u allows one to associate to the a'_i some a''_i in O'', well defined mod$\mathfrak{m}''^e$, and one takes for E'' the extension of F'' obtained by adjoining to F'' a root of the Eisenstein polynomial $X^n + \pi'' \sum a''_i X^i$. The condition $\mathcal{C}^e_{F'}$ ensures that the extension E'' does not depend, up to unique isomorphism, on the choice of the a''_i. Our introduction of the category (ext $Tr_e F)^e$ and of T is a convenient way to verify that one indeed obtains by this construction a functor (ext $F')^e \longrightarrow$ (ext $F'')^e$.

The definition of the category (ext $S)^e$ and that of the functor T will be given in paragraph 2.

1.4. We organize the triples (R, M, ϵ) of 1.1 into a category $\mathcal{T}$ by defining a **morphism** f from $S = (R, M, \epsilon)$ to $S' = (R', M', \epsilon)$ as the data of an integer r, (the **ramification index**), of a morphism $\varphi : R \longrightarrow R'$ and of an R-linear morphism $\eta : M \longrightarrow M'^{\otimes r}$ sending a generator of M onto one of $M'^{\otimes r}$, i.e. inducing an isomorphism, still denoted η, $\eta : M \otimes_R R' \xrightarrow{\sim} M'^{\otimes r}$, and making the following diagram commutative

$$\begin{array}{ccc} M & \xrightarrow{\eta} & M'^{\otimes r} \\ \epsilon \downarrow & & \downarrow \epsilon_{0,r} \\ R & \xrightarrow{\varphi} & R' \end{array} \quad .$$

Let π be a generator of the maximal ideal $\mathfrak{m}$ of R. If $\varphi(\pi) \neq 0$, we necessarily have $r = v(\varphi(\pi))$. If $\varphi(\pi) = 0$, we have $r \geq \lg(R')$. We compose the morphisms by the formula

$$(r', \varphi', \eta') \circ (r, \varphi, \eta) = (r'r, \varphi' \circ \varphi, \eta'^{\otimes r} \circ \eta) \quad .$$

Example 1.4.1. Let F be a local field with perfect residue field, E a finite extension of F, r the ramification index and e an integer. The

structural morphism $F \longrightarrow E$ induces morphisms

$$\varphi : O_F/\mathfrak{m}_F^e \longrightarrow O_E/\mathfrak{m}_E^{re} \quad \text{and} \quad n : \mathfrak{m}_F/\mathfrak{m}_F^{e+1} \longrightarrow \mathfrak{m}_E^r/\mathfrak{m}_E^{r(e+1)} = \mathfrak{m}_E^r/\mathfrak{m}_E^{r+re} \ldots$$

and [1] (r, φ, n) is a morphism $\mathrm{Tr}_e(F) \longrightarrow \mathrm{Tr}_{re}(E)$.

Example 0.1 (1.4.2)**.** Let $S = (R, M, \varepsilon)$ in T, $\mathfrak{m}$ the maximal ideal of R and e an integer such that $1 \leq e \leq \lg R$. The **reduction** mod$\mathfrak{m}^e$ of S is the triple $(R/\mathfrak{m}^e, M/\mathfrak{m}^e M, \varepsilon)$. We have a natural morphism, of ramification index 1, from S to S mod $\mathfrak{m}^e$.

Let $S = (R, M, \varepsilon)$ and $S' = (R', M', \varepsilon)$ in T and $f = (r, \varphi, n) : S \to S'$ a morphism. We denote by $\mathfrak{m}, \mathfrak{m}'$ the maximal ideals of R, R' and k, k' the residue fields.

We will say that f is *flat*, or that S' is *flat* over S if $\lg R' = r \lg R$. If we exclude the case where R is a field ($\lg R = 1$), this amounts to asking that R' be flat over R. The morphisms 1.4.1 are flat.

We will say that S' is *finite* over S if R' is an R-module of finite type, i.e. if the residue field k' of R' (or, as we will say, of S') is a finite extension of that, k, of R. The morphisms 1.4.1 and 1.4.2 are finite.

We will say that S' is *étale* over S if it is flat, finite and $r = 1$. Recall that k is assumed to be perfect and that the residual extension k'/k is therefore separable. If S' is étale over S, R' is the unique étale R-algebra with residue field k'. It is unique up to unique isomorphism. We have $M \otimes_R R' \xrightarrow{\sim} M'$, and ε is obtained by extension of scalars. The functor "residue field" is an equivalence (the S' étale over S) $\longrightarrow$ (the finite extensions of k) and for S' étale over S, with residue field k' and S'' over S, with residue field k'', we have

$$\mathrm{Hom}_S(S', S'') \xrightarrow{\sim} \mathrm{Hom}_k(k', k'') \quad .$$

We will say that S' is *totally ramified* over S if $k \xrightarrow{\sim} k'$. Every finite morphism $f : S \longrightarrow S'$ admits a unique factorization $S \longrightarrow S'_0 \longrightarrow S'$ with S'_0 étale over S, with residue field that of S', and S' totally ramified over S'_0.

1.4.3. Imitating the case of field extensions, we can describe T finite and flat over S and totally ramified in terms of an *Eisenstein polynomial.* Set $S = (R, M, \varepsilon)$, $T = (A, I, \varepsilon)$ and let (r, φ, n) be the morphism $S \longrightarrow T$.

[1] [[UNCERTAIN: $\cdots \otimes \mathfrak{m}_E/\mathfrak{m}_E^{1+re} \ldots | CONFIDENCE : 20 percent : REASON : textcutoff/illegible$]]

Let x be a generator of I. As an R-module, A is free with basis the $\varepsilon(x)^i$ $(0 \leq i < r)$. Since $\eta : M \otimes_R A \to I^{\otimes r}$ is an isomorphism, there exists a unique system of elements $a_i \in M$ such that

$$x^{\otimes r} + \eta\left(\sum_1^r a_i \varepsilon(x)^{r-i}\right) = 0, \tag{1.4.3.1}$$

and, because $x^{\otimes r}$ is a generator of $I^{\otimes r}$, a_r is a generator of M. We call (1.4.3.1) the Eisenstein equation satisfied by x. The a_i allow one to reconstitute T/S, equipped with x: we have

$$A \simeq R[X]/(X^r + \sum \varepsilon(a_i) X^{r-i}),$$

I is the free A-module equipped with a basis x, with $\varepsilon : x \mapsto X$ and η is defined by the condition that its inverse: $I^{\otimes r} \xrightarrow{\sim} M \otimes_R A$ sends $x^{\otimes r}$ to $\sum a_i \otimes X^{r-i}$. Every system of a_i, with a_r a generator of M, thus provides T/S finite flat totally ramified equipped with a generator of I, said to be deduced from S by adjunction of a root of (1.4.3.1).

Si E is the extension of the local field F, with perfect residue field, obtained by adjoining to F a root x of an Eisenstein polynomial $X^r + \sum a_i X^{r-i}$, and if $\bar{a}_i$ is the image of a_i in $\mathfrak{m}/\mathfrak{m}^{e+1}$, $\mathrm{Tr}_{re}(\bar{E})$, equipped with the image of x in $\mathfrak{m}_E/\mathfrak{m}_E^{re+1}$, is deduced from $\mathrm{Tr}_e(F)$ by adjunction of a root of $x^{\otimes r} + \eta(\sum \bar{a}_i \varepsilon(x)^{r-i})$.

Lemma 0.1 (1.4.4)**.** *Let F be a local field with perfect residue field, e an integer ≥ 1, $S = \mathrm{Tr}_e(F)$ and T finite and flat over S, of ramification index r. There exists a finite separable extension E of F, of ramification index r, such that T/S is isomorphic to $\mathrm{Tr}_{re}(E)/\mathrm{Tr}_e(F)$ (1.4.1).*

Proof. Factoring T/S into $T/S'/S$, with S'/S unramified and T/S' totally ramified, we reduce to assuming T/S etale or totally ramified. If T/S is etale, we take for E the unramified extension of F with residue extension that of T/S. If T/S is totally ramified, described by an Eisenstein equation (1.4.3.1), we lift the $a_i \in \mathfrak{m}/\mathfrak{m}^{e+1}$ to $\tilde{a}_i \in \mathfrak{m}$ and we take for E the extension of F obtained by adjoining a root of the Eisenstein polynomial $X^r + \sum \tilde{a}_i X^{r-i}$. For $r \neq 1$, take $\tilde{a}_{r-1} \neq 0$ to ensure that E/F is separable. □

1.5. Let T/S be finite, flat, and totally ramified, and let us resume the notation of 1.4.3. Let us denote by b_i the following element of $I^{\otimes i}$ $(0 \leq i \leq r)$: $b_o = 1$ and for $i > 0$,

$$b_i = \binom{r}{i} x^{\otimes i} + \sum_{j \leq i} \binom{r-j}{i-j} \varepsilon_{i,r} n(a_j) \varepsilon(x)^{i-j} \ .$$

If $S = \mathrm{Tr}_e(F)$, such that E is the extension of F obtained by adjoining to F a root x of the Eisenstein polynomial $P(X) = X^r + \sum \tilde{a}_i X^{r-i}$ and such that $T = \mathrm{Tr}_{re}(E)$, equipped with the generator of $I = m_E/m_E^{re+1}$ image of x, so that a_i is the reduction of $\tilde{a}_i$, the b_i are obtained by reduction from the coefficients of the polynomial $\sum \tilde{b}_i X^{r-i} := P(X + x)$. We have $b_o = \tilde{b}_o = 1$ and $b_r = \tilde{b}_r = 0$.

Let η_T be the function $[0, r-1] \to \mathbb{R}$ such that the Newton polygon, convex hull of the set of $(i, v(b_i))$, is the set of (x, y) with $y \geq \eta_T(x)$. The function η_T is linear from $r-2$ to $r-1$ and we denote by $\tilde{\eta}_T$ its extension to $[0, r]$ linear from $r-2$ to r.

Proposition 0.2 (1.5.1)**.** *With the preceding notation, if the real number f satisfies $0 \leq f \leq e$, for E/F to satisfy the condition C^f of 1.3, <u>it is necessary and sufficient that $\eta_T(r) < r(f+1)$</u>. If this condition is satisfied, <u>the convex hull of the set of $(i, v(b_i))$ coincides with that of</u> <u>the set of $(i, v(\tilde{b}_i))$</u>.*

<u>Notation</u>. We will use the notation v_E of A.2.a.

Proof. Let η_E be the function $[0, r-1] \to \mathbb{R}$ such that the convex hull of the $(i, v(\tilde{b}_i))$ is the set of (x, y) with $y \geq \eta_E(x)$. The theory of the Newton polygon and (A.3) say that, if z_o is a root of P in a separable extension of F large enough, and if we arrange the other roots in an order such that $v(z_i - z_o)$ is increasing with i $(1 \leq i \leq r-1)$, the function η_E linearly interpolates the function $n \mapsto \sum_{i \leq n} v_E(z_i - z_o)$. Let $\tilde{\eta}_E : [0, r] \to \mathbb{R}$ be the extension of η_E, linear from $r-2$ to r. According to A.6.1, the condition C^f is equivalent to

$$\tilde{\eta}_E(r) < r(f+1) \ .$$

Since $b_i \in I^{\otimes i}$ is the reduction of $\tilde{b}_i \in m_E^i$, we have □

$$v(b_i) = \begin{cases} v(\tilde{b}_i) & \text{if } v(\tilde{b}_i) < re+i \\ \infty & \text{otherwise} \end{cases} .$$

This gives the relation between n_T and n_E: the function n_T is defined by the convex hull of the set of the $(i, v(b_i))$ and n_E by that of the $(i, v(\tilde{b}_i))$. Since $v_E(\tilde{b}_i) \geq i$, the function n_E has slope ≥ 1. If $\tilde{n}_E(r) < r(e+1)$, we therefore have for all $i \leq r$ $\tilde{n}_E(i) < r(e+1) - (r-i)$ and for every extremal point $(i, v(\tilde{b}_i))$ of the convex hull of the $(i, v(\tilde{b}_i))$, $v(\tilde{b}_i) < r(e+1) - (r-i) = re+i$, so that $v(b_i) = v(\tilde{b}_i)$ and $n_T = n_E$. If on the other hand for one of these extremal points we have $v(b_i) = \infty$, i.e. $v(\tilde{b}_i) \geq re+i$, we have for each $j \geq i$, $v(\tilde{b}_j) \geq n(j) \geq re+i+(j-i) = re+j$, $v(b_j) = \infty$ and $\tilde{n}_T(r) = \infty$. This proves the

Lemma 0.3. *1.5.2. The following conditions are equivalent:*
(a) E/F satisfies C^e; (b) $\tilde{n}_E(r) < r(e+1)$; (c) $n_E = n_T$; (d) $\tilde{n}_T(r) \neq \infty$.

Since for $f \leq e$, C^f implies C^e, we deduce 1.5.1.

1.5.3. For t real ≥ -1, let us define r_t by

$$r - r_t = \text{the smallest integer } g \text{ such that } n_T \text{ has slope } < t+1$$

on $[0, g[$. If the equivalent conditions of 1.5.2 are satisfied, this function coincides with that of the same name for E/F (A.3.4). Let us set

$$\varphi_{T/S}(u) = \int_0^u dt/(r_0 : r_t)$$

(and $\varphi_{T/S}(u) = u$ for $-1 \leq u \leq 0$), and let $\Psi_{T/S}(u)$ be the inverse function. If the equivalent conditions of 1.5.2 are satisfied, these are the Herbrand functions φ and Ψ for E/F. These interpretations show that the validity of the conditions of 1.5.2 and, if these are satisfied, the functions r, φ and Ψ do not depend on the choice of the generator x of I.

1.5.4. Let T/S be finite and flat and let us factor T/S into $T/S'/S$ with S'/S étale and T/S' totally ramified. Let us set $S = (R, M, \epsilon)$ and $e = \lg(R)$. For $0 \leq f \leq e$, we will denote by C^f the following condition.

$\mathcal{C}^f$. T/S' satisfies the condition $\tilde{n}_T(r) < r(f+1)$ of 1.5.1. If the condition $\mathcal{C}^e$ is satisfied, we define the functions r_t $(t \geq -1)$, $\varphi_{T/S}$ and $\psi_{T/S}$ as being the functions of the same name for T/S'.

1.6 Let $f : S \to T$ in $\mathcal{T}$, finite and flat. We propose to define a norm morphism $N_{T/S} : T^* \to S^*$. Let us set $S = (R, M, \epsilon)$, $T = (A, I, \epsilon)$, $f = (r, \varphi, n)$ and let n be the degree of the residual extension. As an R-module, the R-algebra A is free of rank nr, whence a norm $N_{A/R} : A^\times \to R^\times$. In degree 0, it is the sought norm.

For x a generator of I, let us define $Nx \in M^{\otimes n}$. Suppose first T étale over S. For π a generator of M and $x = a\eta(\pi)$, we set $N(a\eta(\pi)) = N(a) \cdot \pi^{\otimes n}$. With this definition, $N(x)$ is independent of the choice of π, and for $a \in A^*$ satisfies $N(ax) = N(a)N(x)$. This norm extends to $N : T^* \to S^*$.

Suppose T totally ramified over S. Let (1.4.3.1) be the Eisenstein equation satisfied by x. We set $N(x) = (-1)^r a_r$. For suitable F, we have $S \simeq \mathrm{Tr}_e(F)$ and according to 1.4.4 we can find E/F such that T is $\mathrm{Tr}_{re}(E)$. The norm $I^* \to M^*$ is induced by $N_{E/F}$. In particular, it satisfies $N(ax) = N(a)N(x)$ for $a \in A^*$, therefore it extends by multiplicativity to $N : T^* \to S^*$.

For arbitrary T/S, $N_{T/S}$ is defined by composition, starting from the totally ramified and étale cases. For a morphism 1.4.1, the diagram

$$\begin{array}{ccc} E^* & \xrightarrow{N} & F^* \\ \downarrow & & \downarrow \\ (\mathrm{Tr}_{re}E)^* & \xrightarrow{N} & (\mathrm{Tr}_e F)^* \end{array}$$

is commutative.

1.7 Analogous arguments provide a <u>trace</u> morphism

$$\mathrm{Tr} : I \longrightarrow M .$$

For T étale over S, and π a generator of M, it is deduced from $\mathrm{Tr}_{A/R}$ by $\mathrm{Tr}(x\eta(\pi)) = \mathrm{Tr}_{A/R}(x) \cdot \pi$. For T totally ramified over S, and x a generator of I, we take $\mathrm{Tr}(x) = -a_1$ (a_i as in 4.3.1). For arbitrary x in I, we write x as a sum of generators: $x = \Sigma x_i$,

and we set $\mathrm{Tr}(x) = \Sigma\,\mathrm{Tr}(x_i)$. That the trace thus defined does not depend on the choice of the x_i follows from the fact that, for $f : S \longrightarrow T$ of type I.4.1, Tr is induced by $\mathrm{Tr}_{E/F}$. The general case is obtained by composition, and we have: for T/S defined by an extension E/F (1.4.1), the diagram

$$\begin{array}{ccc} m_E & \xrightarrow{\mathrm{Tr}_{E/F}} & m_F \\ \downarrow & & \downarrow \\ I & \xrightarrow{\mathrm{Tr}} & M \end{array}$$

is commutative.

2. THE MAIN THEOREM

2.1 Let $S = (R, M, \varepsilon)$ be a triple as in 1.1, i.e. an object of T (1.4). We define the category (ext S) as being the category of S' over S, i.e. equipped with $f : S \longrightarrow S'$, with S' finite and flat over S. A morphism $g : (S', f') \longrightarrow (S'', f'')$ is a morphism $g : S' \longrightarrow S''$ such that $gf' = f''$. Automatically, S'' is flat over S': if $S' = (A', I', \varepsilon)$, $S'' = (A'', I'', \varepsilon)$, $f' = (r', \varphi', n')$, $f'' = (r'', \varphi'', n'')$ and $g = (r, \varphi, n)$, we have $r'' = rr'$, because $f'' = gf'$, and from $\lg A' = r' \lg R$ and $\lg A'' = r'' \lg R$ it results that $\lg A'' = r \lg A'$.

Let F be a local field with perfect residue field and e an integer ≥ 1. We define a functor

$$T_o : (\text{finite extensions of } F) \longrightarrow (\mathrm{ext}\,\mathrm{Tr}_e\, F)$$

by attaching to an extension E of F, of ramification index r, the triple $\mathrm{Tr}_{er}(E)$, equipped with the morphism 1.4.1: $\mathrm{Tr}_e\, F \longrightarrow \mathrm{Tr}_{er}\, E$. For $u : E \longrightarrow E'$ a morphism of extensions, of ramification index s, $T_o(u) : \mathrm{Tr}_{er}(E) \longrightarrow \mathrm{Tr}_{ers}(E')$ is the morphism 1.4.1 deduced from u.

Lemma 0.4. *2.1.1 (i) The functor T_o is essentially surjective on the isomorphism classes of objects.*

(ii) For E'/E to be étale (resp. totally ramified), it is necessary and sufficient that $T_o(E')/T_o(E)$ be so.

(iii) If $f \leq e$, for $T_o(E)$ to satisfy C^f (1.5.4), it is necessary and sufficient

that E satisfies C^f (1.3).

Assertion (ii) is trivial, (i) is (1.4.4) and (iii) reduces to the totally ramified case, treated in 1.5.1.

2.2.

Let X', X'' be in (ext S). Notations: $X' = (A', I', \epsilon)$, $(r', \varphi', \eta') : S \longrightarrow X'$ is the structural morphism and similarly for X''. We suppose X' totally ramified. Let x be a generator of I' and

$$x^{\otimes r'} + \eta'(\sum a_i \epsilon(x)^{r'-i}) = 0 \tag{2.2.1}$$

its Eisenstein equation (1.4.3). There is a morphism from X' to X'' only if $r' \mid r''$. Suppose that such is the case and set $s = r''/r'$. We deduce from the description 1.4.3 of X as deduced from S by adjunction of a root of (2.2.1) that the data of a morphism from X' to X'' is equivalent to that of $y \in I''^{\otimes s}$ (the image of x) satisfying

$$y^{\otimes r'} + \eta''(\sum a_i \epsilon_{os}(y)^{r'-i}) = 0 \quad \text{in} \quad I''^{\otimes r''} \xrightarrow[\sim]{\eta''} M \otimes_R A''. \tag{2.2.2}$$

For $S = \mathrm{Tr}_e(F)$, $X' = \mathrm{T}_o(E')$, $X'' = \mathrm{T}_o(E'')$, x the image of $\tilde{x}$ satisfying the Eisenstein equation $\tilde{x}^{r'} + \sum \tilde{a}_i \tilde{x}^{r'-i} = 0$ and $y \in I''^{\otimes s}$ image of $\tilde{y} \in \mathfrak{m}''^s$, for 2.2.2 to be satisfied, it is necessary and sufficient that

$$v_F(\tilde{y}^{r'} + \sum \tilde{a}_i \tilde{y}^{r'-i}) \geq e + 1. \tag{2.2.3}$$

One has indeed $I''^{\otimes r''} \approx \mathfrak{m}''^{r''}/\mathfrak{m}''^{r''(e+1)}$.

2.3.

Let X' and X'' be in (ext S) and let us resume the notations of 2.2. For f a real number ≥ 0, we define $R(f)$ as being the following equivalence relation between morphisms from X' to X'': if $u_i : X' \to X''$ $(i = 1, 2)$ are two morphisms from X' to X'', $u_i = (s, \varphi_i, \eta)$, $u_1 \equiv u_2 \pmod{R(f)}$ signifies that we have: $R(f)$. φ_1 and φ_2 induce the same morphism on the residue fields and, for all x in I', we have $v(\eta_1(x) - \eta_2(x)) \geq s(f+1)$.

Remarks (i) If X' is totally ramified over S and x_0 is a generator of I', we have

$$u_1 \equiv u_2 \pmod{R(f)} \iff v(\eta_1(x_0) - \eta_2(x_0)) \geq r(f+1) \ .$$

(ii) Let k, k' and k'' be the residue fields of S, X' and X'', and X'_{nr} the etale extension of S with residual extension k'/k. We have

$$\mathrm{Hom}_S(X'_{nr}, X'') \xrightarrow{\sim} \mathrm{Hom}_k(k', k'') \quad (1.4.2).$$

If $\bar{\varphi} : k' \to k''$ is given, whence $\varphi_{nr} : X'_{nr} \to X''$, we have

$$\mathrm{Hom}_{X'_{nr}}(X', X'') \xrightarrow{\sim} \{u \in \mathrm{Hom}_S(X', X'') \mid u \text{ induces } \bar{\varphi}\}.$$

For two morphisms on the left-hand side to be $R(f)$-equivalent, it is necessary and sufficient that their images be so.

Lemma 0.5 (2.3.1)**.** *Let $S = Tr_e(F)$, $X' = T_o(E')$ and $X'' = T_o(E'')$. Suppose that E' satisfies C^e and set $e' = \psi_{E'/F}(e) = \psi_{X'/S}(e)$ (1.5.3). The functor T_o induces a bijection of* $\mathrm{Hom}_F(E', E'')$ *with the set of $R(e')$-equivalence classes of morphisms $X' \to X''$ in* (ext S)*.*

Proof. We retain the notation of 2.2 and 2.3. Suppose first of all that E'/F is totally ramified. Let $\tilde{x}$ be a uniformizer of E', x its image in I' and P the Eisenstein polynomial of which $\tilde{x}$ is a root. According to 2.2, giving $u : X' \to X''$ amounts to giving $\tilde{y}$ in E'', of valuation the quotient $s = r''/r'$ of the ramification indices of X' and X'', modulo elements of valuation $\geq s + er''$ and satisfying (2.2.3). The condition C^e guarantees that for y_1, y_2 in E'' roots of P, we have $v_{E''}(y_1 - y_2) > \psi_{E'/F}(e) + 1$ and that for $\tilde{y}$ satisfying (2.2.3), there exists one (and only one) root $\tilde{y}_1$ with $v_{E''}(\tilde{y} - \tilde{y}_1) \leq \psi_{E'/F}(e) + 1$. For $\tilde{y}_1$ and $\tilde{y}_2$ satisfying (2.2.3), that the corresponding morphisms be congruent mod $R(f)$ is equivalent to

$$r' v_F(\tilde{y}_1 - \tilde{y}_2) \leq f + 1.$$

The assertion follows for E'/F totally ramified, and we deduce the general case by remark (ii) above. □

Lemma 0.6 (2.4)**.** *Let X', X'' and X''' be in* (ext S)*, $u : X'' \to X'''$ and $v_1, v_2 : X' \to X''$. For any f, if $v_1 \equiv v_2 \pmod{R(f)}$, then $uv_1 \equiv uv_2 \pmod{R(f)}$.*

The verification is left to the reader.

Lemma 0.7. *Let X', X'' and X''' be in* (ext S)*, $u_1, u_2 : X'' \to X'''$ and $v : X' \to X''$. Assume that X' and X'' satisfy $C^{\widehat{e}}$. If $u_1 \equiv u_2 \pmod{R(\Psi_{X''/S}(e))}$, then $u_1 v \equiv u_2 v \pmod{R(\Psi_{X'/S}(e))}$.*

Proof. Suppose first that X' and X'' are totally ramified over S. We can assume (1.2), and we suppose that $S = \mathrm{Tr}_e(F)$, then that $X' = T_o(E')$, $X'' = T_o(E'')$ and $X''' = T_o(E''')$. According to 2.3.1, there exists $j : E' \to E''$ such that $v \equiv T_o(j) \pmod{R(\Psi_{E'/F}(e))}$. Applying 2.4, we reduce to assuming that $v = T_o(j)$. Let x be a uniformizer of E', y one of E'' and identify E' with a subfield of E'', using j. Write x as a polynomial in y with coefficients in O_F: $x = G(y)$, and let P and Q be the Eisenstein polynomials with coefficients in F of which x and y are respectively roots.

According to 2.2, each morphism $X'' \to X'''$ is defined by an approximate solution y' of Q in $E''' : v_{E'''}Q(y') \geq e + 1$. For two approximate solutions y', y'' to define morphisms congruent mod $R(\Psi_{E''/F}(e))$, it is necessary and sufficient that

$$v_{E'''}(y' - y'') \geq \Psi_{E''/F}(e) + 1.$$

We have an analogous description for the morphisms $X' \to X'''$, in terms of approximate solutions of P, and the composition with $v : X' \to X''$ is interpreted as $y' \mapsto G(y')$. This reduces 2.5 (totally ramified case) to □

Lemma 0.8. *If y' and y'' are approximate solutions of Q: $v_F Q(y'), v_F Q(y'') \geq e+1$, and $v_{E'''}(y''-y') \geq \Psi_{E''/F}(e)+1$, then*

$$v_{E'}(G(y'')-G(y')) \geq \Psi_{E'/F}(e)+1.$$

Let R be the minimal polynomial of y over E'. Since $x = G(y)$, we have $R(X) \mid x - G(X)$. Applying 2.3.1, we reduce to assuming that one of the approximate solutions y', y'' – say y'' – is an exact solution : $Q(y'') = 0$. To this solution corresponds an embedding $E'' \hookrightarrow E'''$, by which we will identify E'' with a subfield of $E''' : y = y''$. With these notations, we must verify that if y' in E''' satisfies $v_{E''}(y'-y) \geq \Psi_{E''/F}(e)+1$, then

$$v_{E'}(x - G(y')) \geq \Psi_{E'/F}(e)+1.$$

Since $R(X) \mid x - G(X)$, it suffices to establish that

$$v_{E'} R(y') \geq \Psi_{E'/F}(e)+1 \ .$$

We have $\Psi_{E''/F} = \Psi_{E''/E'} \circ \Psi_{E'/F}$. Let $e' = \Psi_{E'/F}(e)$. The extension E'' of E' satisfies $C^{e'}$ and we must check that

$$v_{E''}(y'-y) \geq \Psi_{E''/E'}(e')+1 \Rightarrow v_{E'} R(y') \geq e'+1 \ .$$

This follows from A.6.4, applied to the extensions E'', E''' of E' .

Let us prove 2.5 in the general case. Replacing X' by its étale extension with residue field the residue field of X'' , we reduce to assuming that X' and X'' have the same residue field k' (use 2.3 remark (ii)). Replacing S by its étale extension of residual extension k' (new application of 2.3 remark (ii)), we reduce to the totally ramified case already treated.

Definition 0.1 (2.7)**.** Let $S = (R, M, \epsilon)$ with $lg\ R = e$. The category (ext $S)^e$ is the category of objects whose objects are the objects of (ext S) satisfying C^e , a morphism from X to Y being an $R(\psi_{X/S}(e))$-equivalence class of (ext S)-morphism from X to Y . The composition of morphisms is obtained by passing to the quotient from the composition of morphisms in (ext S) .

Lemmas 2.4 and 2.5 guarantee that the given definition of the composition of morphisms makes sense. We have a quotient functor

$$\text{(full subcategory of (ext } S) \text{ formed by the objects satisfying } C^e) \to (\text{ext } S)^e \ .$$

For $S = \mathrm{Tr}_e(F)$, composing it with the restriction of T_o to (ext $F)^e$, we obtain a functor

$$T : (\text{ext } F)^e \longrightarrow (\text{ext } \mathrm{Tr}_e F)^e \ .$$

Lemma 2.3.1 and 2.1.1 (iii) for $f = e$ provide our main theorem :

Theorem 0.9 (2.8)**.** *The functor $T : (\text{ext } F)^e \longrightarrow (\text{ext } \mathrm{Tr}_e F)^e$ is an equivalence of categories.*

Here are some immediate corollaries.

Corollary 0.10 (2.9)**.** *<u>Let</u> $S = (R, M, \varepsilon)$ in T, with $\lg(R) = e$, f an <u>integer</u>, $1 \leq f \leq e$, and $\overline{S}$ the reduction of $S \mod \mathfrak{m}^f$ (1.4.2). <u>The functor</u> which to S'/S, of ramification index r, associates its reduction $\mod \mathfrak{m}_{S'}^{rf}$ <u>induces an equivalence</u>*

$$(\textit{objects of } (\text{ext } S)^e, \ \textit{satisfying } C^f) \longrightarrow (\text{ext } \overline{S})^f.$$

Proof. We write $S = \mathrm{Tr}_e(F)$. The functor T induces an equivalence

$$(\text{ext } F)^f \longrightarrow (\text{objects of } (\text{ext } \mathrm{Tr}_e F)^e \text{ satisfying } C^f),$$

and 2.9 is the composite of its inverse with the equivalence 2.8 for (ext $F)^f$:

$$(\text{ext } F)^f \longrightarrow (\text{ext } \overline{S})^f.$$

□

Remark 1. Using that, in (2.3.1), we do not assume that the target E'' satisfies C^e, we can strengthen 2.9 as follows:

(a) Any $\overline{S}'/\overline{S}$ in (ext $\overline{S}$) is the reduction of an S'/S in (ext S). If $\overline{S}'$ satisfies C^f, S' also (use 2.1.1 (i) and (iii)).

(b) Let S' and S'' be in (ext S), of reduction $\overline{S}'$ and $\overline{S}''$. If S' satisfies C^f, we have

$$\mathrm{Hom}_{\text{ext } S}(S', S'') \mod R(\Psi_{S'/S}(e)) \xrightarrow{\sim} \mathrm{Hom}_{\text{ext } \overline{S}}(\overline{S}'/\overline{S}'') \mod R(\Psi_{S'/S}(f)).$$

2.10. Let E in (ext F) satisfying C^e. For an extension E' of E to satisfy C^e (over F), it is necessary and sufficient that as an extension of E it satisfies $C^{\Psi_{E/F}(e)}$:

$$(\text{ext } \underline{F})^{\Psi_{E/F}(e)}_{E/F} \xrightarrow{\sim} (\text{ext } F)^e/E.$$

Translating by 2.8, we obtain the

Corollary 0.11 (2.10.1)**.** *<u>Let</u> $S = (R, M, \varepsilon)$ in T, with $\lg R = e$, and T in* (ext S) *<u>satisfying</u> C^e, of ramification index r over S. <u>The</u> evident functor is an equivalence*

$$(\textit{objects of } (\text{ext } T)^{re} \textit{ satisfying } C^{\Psi_{T/S}(e)}) \longrightarrow (\text{ext } S)^e/T.$$

Let T' and T'' in (ext T)*, T' and T'' satisfying the condition $C^{\Psi_{T/S}(e)}$*

$$\begin{tikzcd} S \arrow[r] & T \arrow[r, "u"] \arrow[dr, "v"'] & T' \\ & & T'' \end{tikzcd}$$

The corollary signifies that we have a bijection:

$$(\text{classes mod } R(\Psi_{T'/T}(re)) \text{ of T-morphisms from } T' \text{ to } T'') \xrightarrow{\sim}$$

$$(\text{classes mod } R(\Psi_{T'/S}(e)) \text{ of S-morphisms } f \text{ from } T' \text{ to } T'', \text{ such that } fu \equiv v \mod R(\Psi_{T/S}(e))).$$

Let us take $T' = T''$, $u \equiv v \mod R(\Psi_{T/S}(e))$ and $f = \mathrm{Id}$. We find that there exists an automorphism a of T' with $a \equiv \mathrm{Id} \mod R(\Psi_{T'/S}(e))$ such that $au = v$:

Corollary 0.12 (2.10.2)**.** *Let T and T' be in* (ext$S$)*, satisfying $\mathcal{C}^e$ and $u, v : T \longrightarrow T'$ two S-morphisms. For $u \equiv v \mod R(\Psi_{T/S}(e))$, it is necessary and sufficient that there exists an S-automorphism a of T', $a \equiv Id \mod R(\Psi_{T'/S}(e))$, such that $au = v$.*

The sufficiency is a particular case of 2.5.

3. COMPATIBILITIES

3.1. Let $S = (R, M, c)$ in $\mathcal{T}$, with $\lg R = e$. The definition given of the category (ext$S)^e$ has this unpleasant feature, that the arrows are defined as equivalence classes of "true arrows" (in (ext$S$)). Various objects attached to T in (ext$S)^e$ are however defined up to unique isomorphism, and functorial in T.

For $T = (A, I, c)$ in (ext$S$), satisfying $\mathcal{C}^e$, let $\mathfrak{m}_T$ be the maximal ideal of A and set

$$T_1 := \text{reduction of } T \mod \mathfrak{m}_T^{\Psi_{T/S}(e)}. \tag{1.4.2}$$

Recall that $\Psi_{T/S}(e)$ is an integer (A. 5.4). For any morphism $u : T' \longrightarrow T''$ between objects satisfying $\mathcal{C}^e$, of ramification index s, we have

$$\Psi_{T''/S}(e) = \Psi_{T''/T'}\Psi_{T'/S}(e) \leq s\Psi_{T'/S}(e) \text{ and } u \text{ induces a morphism in}$$

$T'/S : T'_1 \longrightarrow T''_1$. If $u \equiv v \mod \mathfrak{R}(\Psi_{T'/S}(e))$, the morphisms from T'_1 to T''_1 deduced from u and v coincide. By passing to the quotient, one obtains a functor

(3.1.1) $\qquad T \longmapsto T_1 \ : \ (\text{ext } S)^e \longrightarrow T/S$.

Proposition 0.13. *Let T' and T'' be in* (ext S)*, satisfying C^e, and $f : T' \longrightarrow T''$.*
(i) The norm morphism 4.3 induces a norm morphism $N : T_1''^* \longrightarrow T_1'^*$.
(ii) This morphism depends only on the $\mathfrak{R}(\Psi_{T'/S}(e))$-equivalence class of f.

Proof. For (i), replacing S by T' and e by $\Psi_{T'/S}(e)$, we reduce to assuming that $S = T'$. Let $T = T''$. One can assume that, for a suitable extension E/F, of ramification index r, one has $S = \mathrm{Tr}_e(F)$ and $T = \mathrm{Tr}_{re}(E)$. The norm morphism 1.6 is then induced by $N_{E/F} : E^* \longrightarrow F^*$. Let U^n denote the group of units congruent to 1 modulo the n-th power of the maximal ideal. It remains to use that $N_{E/F}$ sends $U^{\Psi_{E/F}(e)}(E)$ into $U^e(F)$.

Let us prove (ii). By 2.10.2 and the transitivity of norms, it suffices to observe that an automorphism a of T'', whose image is the identity automorphism of T'' in (ext $S)^e$, induces the identity automorphism of T_1''. □

Proposition 3.2 allows one to attach to each morphism $f : T' \longrightarrow T''$ in (ext $S)^e$ a norm morphism

(3.2.1) $\qquad N : (T_1'')^* \longrightarrow (T_1')^* \quad .$

For traces, we have similarly

Proposition 0.14. *Let T' and T'' be in* (ext S) *satisfying C^e and $f : T' \longrightarrow T''$. Let $T_1' = (A_1', I_1', \epsilon)$ and $T_1'' = (A_1'', I_1'', \epsilon)$. The trace morphism 1.7 induces a trace morphism :* $\mathrm{Tr} : I_1'' \longrightarrow I_1'$*. It depends only on the $\mathfrak{R}(\Psi_{T'/S}(e))$-equivalence class of f.*

One reasons as in 3.2, using that $\mathrm{Tr}_{E/F}$ sends $\mathfrak{m}_E^{\Psi_{E/F}(e)+1}$ into (in fact onto) $\mathfrak{m}_F^{e+1}$.

The proposition allows one to attach to each morphism $f : T' \longrightarrow T''$ in (ext $S)^e$ a trace morphism

(3.3.1)

$$\mathrm{Tr} : \mathrm{I}_1^{''} \longrightarrow \mathrm{I}_1'$$

.

3.4. Let F' and F'' be two local fields, e an integer ≥ 1, and u an isomorphism $\mathrm{Tr}_e(F') \xrightarrow{\sim} \mathrm{Tr}_e(F'')$. We use it to identify $\mathrm{Tr}_e(F')$ and $\mathrm{Tr}_e(F'')$ with the same triple $S = (R, M, \epsilon)$. According to the theorem, u provides an equivalence of categories

$$U : (\mathrm{ext}\, F')^e \xrightarrow{\sim} (\mathrm{ext}\, F'')^e \quad .$$

Construction 3.4.1. Let E' be an extension of F' satisfying $\mathcal{C}_{F'}^e$, and E'' the extension of F'' corresponding to it by U. They have the same ramification index r and $\Psi_{E'/F'} = \Psi_{E''/F''}$. Let $e(1) = \Psi_{E'/F'}(e) = r(e+1) - (\text{valuation of the different}) - 1$ (A.5.4); it is an integer. We construct a canonical isomorphism

$$u : \mathrm{Tr}_{e(1)}(E') \xrightarrow{\sim} \mathrm{Tr}_{e(1)}(E'') \quad .$$

By definition, we have a canonical isomorphism, in $(\mathrm{ext}\, S)^e$, between $T(E')$ and $T(E'')$, i.e. a class mod $R(e(1))$ of isomorphisms between $\mathrm{Tr}_{re}(E') = T_o(E')$ and $\mathrm{Tr}_{re}(E'') = T_o(E'')$. These isomorphisms all induce the same isomorphism between the quotients $\mathrm{Tr}_{e(1)}(E')$ and $\mathrm{Tr}_{e(1)}(E'')$ of $\mathrm{Tr}_{re}(E')$ and $\mathrm{Tr}_{re}(E'')$. This is the desired isomorphism.

3.4.2. We have $e \leq e(1) \leq re$, with $e(1) = re$ if and only if the extension E'/F' is tamely ramified. If $e(1) < re$, we can specify $\tilde{u}$ as a canonical isomorphism of R-algebras

$$(3.4.2.1) \qquad O'/\mathfrak{m}'^{e(1)+1} \xrightarrow{\sim} O''/\mathfrak{m}''^{e(1)+1} \quad .$$

It is a matter of verifying that two isomorphisms $f_1, f_2 : \mathrm{Tr}_{re}(E') \to \mathrm{Tr}_{re}(E'')$ which are $R(e(1))$-equivalent induce the same isomorphism (3.4.2.1). If R_o is the ring of integers of the étale extension of R with residue field that of $\bar{E}'$ and $\bar{E}''$, we have

$$R_o \subset O'/\mathfrak{m}'^{re} \quad , \quad O''/\mathfrak{m}''^{re} \quad ,$$

and each isomorphism f_i induces the identity of R_o: we have $f_1|_{R_o} = f_2|_{R_o}$. For x in the maximal ideal, image of x_1 in $\mathfrak{m}'/\mathfrak{m}'^{re+1}$, we have

$$v(f_1(x) - f_2(x)) \geq v(f_1(x_1) - f_2(x_1)) \geq e(1) + 1 \quad .$$

In total, f_1 and f_2 satisfy $v(f_1 - f_2) \geq e(1) + 1$, thus induce the same morphism (3.4.2.1).

The isomorphism 3.4.1 obeys the following functorialities.

3.4.3. Let $v : E'_1 \longrightarrow E'_2$ be a morphism between extensions of F', in (ext $F')^e$, and $E''_1 \longrightarrow E''_2$ its image under U. Set $e(i) = \Psi_{E'_i/F'}(e) = \Psi_{E''_i/F''}(e)$. If r is the ramification index of E'_2/E'_1, we have $e(2) \leq re(1)$, whence arrows deduced from 1.4.1: $\mathrm{Tr}_{e(1)}E'_1 \longrightarrow \mathrm{Tr}_{e(2)}E'_2$. Similarly for E''_i. The diagram

$$\begin{array}{ccc} \mathrm{Tr}_{e(1)}(E') & \longrightarrow & \mathrm{Tr}_{e(1)}(E'') \\ \downarrow & & \downarrow \\ \mathrm{Tr}_{e(2)}(E') & \longrightarrow & \mathrm{Tr}_{e(2)}(E'') \end{array}$$

is commutative. An analogous result holds for the arrows 3.4.2, insofar as this makes sense.

3.4.4. The canonical isomorphism $\tilde{u}$ of 3.4.1 induces an isomorphism

$$E'^*/U^{e(1)} \simeq \mathrm{Tr}_{e(1)}(F')^* \xrightarrow{\sim} \mathrm{Tr}_{e(1)}(E'')^* = E''^*/U^{e(1)}$$

With the notations (a), the norm N sends $U^{e(2)}(E'_2)$ into $U^{e(1)}(E'_1)$, similarly for E''_2/E''_1, and the diagram

$$\begin{array}{ccc} E'^*_2/U^{e(2)} & \xrightarrow{\sim} & E''^*_2/U^{e(2)} \\ N\downarrow & & \downarrow \\ E'^*_1/U^{e(1)} & \xrightarrow{\sim} & E''^*_1/U^{e(1)} \end{array}$$

is commutative.

3.4.5. Let us denote by $\mathfrak{m}$, decorated with indices and ' or ", the maximal ideal of the valuation ring of E, decorated similarly. The diagram

$$\begin{array}{ccc} \mathfrak{m}'_2/\mathfrak{m}'^{e(2)+1}_2 & \xrightarrow{\sim} & \mathfrak{m}''_2/\mathfrak{m}''^{e(2)+1}_2 \\ \mathrm{Tr}_{E'_2/E'_1}\downarrow & & \downarrow \mathrm{Tr}_{E''_2/E''_1} \\ \mathfrak{m}'_1/\mathfrak{m}'^{e(1)+1}_1 & \xrightarrow{\sim} & \mathfrak{m}''_1/\mathfrak{m}''^{e(1)+1}_1 \end{array}$$

is commutative.

Assertions 3.4.4 and 3.4.5 result respectively from 1.6 and 1.7, and from the definitions.

3.5. The equivalence of categories U induces an isomorphism – unique up to inner automorphism –

$$\mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^{\mathfrak{e}} \simeq \mathrm{Gal}(\overline{F}''/F'')/\mathrm{Gal}(\overline{F}''/F'')^{\mathfrak{e}}, \tag{3.5.1}$$

for $\overline{F}'$ (resp. $\overline{F}''$) a separable closure of F' (resp. F'').

Indeed

(a) It extends into an equivalence of the category of Ind-objects of (ext $F')^{\mathfrak{e}}$ with that of the ind-objects of (ext $F'')^{\mathfrak{e}}$. These categories of Ind-objects are interpreted, concretely, as those of the algebraic extensions of F' (resp. F'') all of whose finite sub-extensions are in (ext $F')^{\mathfrak{e}}$ (resp. (ext $F'')^{\mathfrak{e}}$).

(b) There exists in Ind(ext $F')^{\mathfrak{e}}$ an object Ω', unique up to (non-unique) isomorphism, such that for all E' in (ext $F')^{\mathfrak{e}}$, $\mathrm{Hom}(E', \Omega') \neq \emptyset$. If $\overline{F}'$ is a separable closure of F', one can take

$$\Omega' = \text{invariants of } \mathrm{Gal}(\overline{F}'/F')^{\mathfrak{e}} \text{ in } \overline{F}'.$$

(c) We have $\mathrm{Aut}\ \Omega' = \mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^{\mathfrak{e}}$[2], and 3.5.1 is

$$U : \mathrm{Aut}\ \Omega' \xrightarrow{\sim} \mathrm{Aut}(U\Omega'),$$

the non-uniqueness arising from the fact that Ω' is only unique up to non-unique isomorphism.

By passing to the abelianized groups, (3.5.1) yields a canonical isomorphism

$$(\mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^{\mathfrak{e}})^{\mathrm{ab}} \simeq (\mathrm{Gal}(\overline{F}''/F'')/\mathrm{Gal}(\overline{F}''/F'')^{\mathfrak{e}})^{\mathrm{ab}} \tag{3.5.2}$$

Note that the groups (3.5.2) are independent, up to unique isomorphism, of the choices of $\overline{F}'$ and $\overline{F}''$.

3.6. Suppose F' and F'' have finite residue fields. Class field theory then provides a canonical isomorphism

$$(\mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^{\mathfrak{e}})^{\mathrm{ab}} \simeq (F'^*/U^{\mathfrak{e}}(F'))^{\wedge}$$

[2] [[UNCERTAIN: F' in the denominator of the Galois group is inferred from context and eq (3.5.1); image text is slightly ambiguous but likely F'.]]

($\hat{}$ for "profinite completion").

Proposition 0.15. *The diagram*

$$\begin{array}{ccc} (\mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^e)^{\mathrm{ab}} & \underset{\cong}{\overset{U}{\longrightarrow}} & (\mathrm{Gal}(\overline{F}''/F'')/\mathrm{Gal}(\overline{F}''/F'')^e)^{\mathrm{ab}} \\ \uparrow & & \uparrow \\ (F'^*/U^e)^\wedge \simeq (\mathrm{Tr}_e F')^{*\wedge} & \underset{\cong}{\overset{u}{\longrightarrow}} & \mathrm{Tr}_e(F'')^{*\wedge} \simeq (F''^*/U^e)^\wedge \end{array}$$

is commutative.

The class field theory isomorphism is the projective limit, over the finite abelian extensions $E \subset \overline{F}'$ of F', of the reciprocity isomorphisms

$$\mathrm{Gal}(E/F') \longleftarrow F'^*/\mathrm{N}E^* \quad .$$

It suffices to show that for each abelian extension E' of F' satisfying $C^e_{F'}$, corresponding to an abelian extension E'' of F'', the diagram in solid lines

$$\begin{array}{ccc} \mathrm{Gal}(E'/F') & \xrightarrow{\sim} & \mathrm{Gal}(E''/F'') \\ \uparrow & & \uparrow \\ F'^*/\mathrm{N}E'^* & \dashleftarrow & F''^*/\mathrm{N}E''^* \\ \uparrow & & \uparrow \\ \mathrm{Tr}_e(F')^* & \xrightarrow{\sim} & (\mathrm{Tr}_e(F''))^* \end{array}$$

can be completed into a commutative diagram as indicated. We will verify this using the description of the reciprocity isomorphism provided by Dwork's theorem ([6] XIII §5).

For an abelian extension not necessarily totally ramified E/F, this description is as follows. Let k be the residue field of F, $\overline{k}$ an algebraic closure of k and F_{nr} the completion of the maximal unramified extension of F with residue field $\overline{k}$ (extension of the residue field from k to $\overline{k}$). Let $q := \#k$, φ the F-automorphism of F_{nr} inducing on $\overline{k}$ the automorphism $x \longmapsto x^q$ and Fr the inverse of φ. It is the "Frobenius

...geometric". Let $E_{nr} = E \otimes_F F_{nr}$. It is a product of fields E_i indexed by the k-embeddings into $\bar{k}$ of the residue field of E. We still denote by Fr the automorphism $x \otimes y \longmapsto x \otimes \mathrm{Fr}(y)$ of E_{nr}, and we let $\mathrm{Gal}(E/F)$ act on E_{nr} by $s(x \otimes y) = s(x) \otimes y$. These actions commute.

Let $v_i : E_i^* \to \mathbb{Z}$ be the valuation of E_i and $v : E_{nr}^* = \prod E_i^* \to \mathbb{Z}$ be the sum of the v_i. The sequence

$$0 \longrightarrow E^* \longrightarrow E_{nr}^* \xrightarrow{\mathrm{Fr}(x)/x} E_{nr}^* \xrightarrow{v} \mathbb{Z} \longrightarrow 0$$

is exact, and the first three terms of the analogous exact sequence for F are deduced from it by passing to invariants under $\mathrm{Gal}(E/F)$: we have a commutative diagram with exact rows, where the vertical arrows (except the last one) are the norm morphisms:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E^* & \longrightarrow & E_{nr}^* & \longrightarrow & E_{nr}^* & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & F^* & \longrightarrow & F_{nr}^* & \longrightarrow & F_{nr}^* & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

Let $\pi \in E_{nr}^*$ such that $v(\pi) = 1$. For any $s \in \mathrm{Gal}(E/F)$, we have $v(\pi^{s-1}) = v(\pi^s) - v(\pi) = 0$ and there exists $y \in E_{nr}^*$ such that $\pi^{s-1} = y^{\mathrm{Fr}-1}$. We have $Ny \in F^* \subset F_{nr}^*$ (norm from E_{nr}^* to F_{nr}^*). Indeed, $(Ny)^{\mathrm{Fr}-1} = N(y^{\mathrm{Fr}-1}) = N(\pi^{s-1}) = 1$. Since y is unique $\pmod{E^*}$, the class of $N(y)$ $\pmod{N_{E/F}(E^*)}$ is independent of the choice of y. It is also independent of the choice of π: if $\pi' = \pi x$ with $v(x) = 0$, hence $x = z^{\mathrm{Fr}-1}$, we have $\pi'^{s-1} = (y \cdot z^{s-1})^{\mathrm{Fr}-1}$ and $N(y \cdot z^{s-1}) = N(y) \cdot N(z^{s-1}) = N(y)$. We have: by the reciprocity map, s corresponds to the class $\pmod{N_{E/F}(E^*)}$ of Ny^{-1}. For the proof, see [6] XIII §5, and exercise 2. Note two sign changes with respect to [6]: we use the geometric Frobenius, and we normalize the class field isomorphism so that it transforms geometric Frobenius elements into uniformizers.

Let $e' = \psi_{E/F}(e)$. The above description of the image of s by the reciprocity map in the quotient F^*/NE^* of $F^*/U^e(F)$ depends only on (a) the $E_i^*/U^{e'}(E_i)$, for $E_{nr} = \prod E_i$. We set $U^{e'}(E_{nr}) = \prod U^{e'}(E_i)$. (b) The commutative diagram with exact rows

$$\begin{array}{ccccccccc} 0 & \longrightarrow & E^*/U^{e'}(E) & \longrightarrow & E_{nr}^*/U^{e'}(E_{nr}) & \longrightarrow & E_{nr}^*/U^{e'}(E_{nr}) & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \| & & \\ 0 & \longrightarrow & F^*/U^e(F) & \longrightarrow & F_{nr}^*/U^e(F_{nr}) & \longrightarrow & F_{nr}^*/U^e(F_{nr}) & \longrightarrow & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

of vertical arrows induced by the norm; (c) the automorphism induced by s of $E_{nr}^*/U^{e'}(E_{nr})$. It is a matter of reconstructing these data from $S = \mathrm{Tr}_e(F')$, from T/S in $(\mathrm{ext}\ S)^e$ corresponding to E and from the automorphism s of T. Let k be the residue field of S, k_T that of T, $\bar{k}$ an algebraic closure of k and I the set of k-embeddings of k_T into $\bar{k}$. Let S_{nr} be obtained from S by extension of the residue field to $\bar{k}$ and for $i \in I$, let similarly T_i be obtained from T by the extension of residue field $i : k_T \hookrightarrow \bar{k}$. The $E_i^*/U^{e'}(E_i)$ are the T_{i1}^*. The commutative diagram (b) is

$$\begin{array}{ccccccccc} 0 \longrightarrow & T_1^* & \longrightarrow & \prod T_{o1}^* & \longrightarrow & \prod T_{i1}^* & \longrightarrow & \mathbb{Z} & \longrightarrow 0 \\ & N\downarrow & & \prod N\downarrow & & \prod N\downarrow & & \| & \\ 0 \longrightarrow & S^* & \longrightarrow & S_{nr}^* & \longrightarrow & S_{nr}^* & \longrightarrow & \mathbb{Z} & \longrightarrow 0 \end{array},$$

and the automorphism s of T in $(\mathrm{ext}\ S)^e$ provides in $(\mathrm{ext}\ S_{nr})^e$ isomorphisms $s : T_i \to T_{s(i)}$, and finally the desired automorphism $\prod T_{i1}^* \to \prod T_{i1}^*$ (c). This ends the proof of 3.6.1.

3.7. Let, as in 3.4, F', F'' and $u : \mathrm{Tr}_e(F') \xrightarrow{\sim} \mathrm{Tr}_e(F'')$. The category equivalence U provides an isomorphism

$$\mathrm{Gal}(\bar{F}'/F')/\mathrm{Gal}^e \xrightarrow{\sim} \mathrm{Gal}(\bar{F}''/F'')/\mathrm{Gal}^e,$$

well-defined up to inner isomorphism, whence a natural bijection of the set of isomorphism classes of complex representations (continuous, of finite dimension) of $\mathrm{Gal}(\bar{F}'/F')$, trivial on $\mathrm{Gal}(\bar{F}'/F')^e$, with the analogous set for F''.

Suppose F' and F'' have finite residue fields. One can then above replace the Galois groups with the Weil groups. We will say that two additive characters $\Psi' : F' \longrightarrow \mathbb{C}^*$ and $\Psi'' : F'' \longrightarrow \mathbb{C}^*$ correspond, if the following conditions are verified:

(a) $n(\Psi') = n(\Psi'')$. Recall that $n(\Psi')$ is the largest integer $n \in \mathbb{Z}$ such that Ψ' is trivial on the fractional ideal $\mathfrak{m}'^{-n}$ $(-n)$-th power of the maximal ideal of the valuation ring of F'. Let us set $n = n(\Psi') = n(\Psi'')$.

(b) The isomorphism u identifies $\Psi'|(\mathfrak{m}'^{-n-e}/\mathfrak{m}'^{-n})$ to $\Psi''|(\mathfrak{m}''^{-n-e}/\mathfrak{m}''^{-n})$, via the isomorphism

$$\mathfrak{m}'^{-n-e}/\mathfrak{m}'^{-n} \simeq (\mathfrak{m}'/\mathfrak{m}'^{e+1}, \otimes(-n-e)) \simeq (\mathfrak{m}''/\mathfrak{m}''^{e+1}, \otimes(-n-e)) \simeq \mathfrak{m}''^{-n-e}/\mathfrak{m}''^{-n}\ .$$

We endow F and F'' with Haar measures dx for which the valuation rings of F' and F'' have the same volume.

Proposition 0.16. *With the notation above, if a representation V' of $\mathrm{Gal}(\overline{F}'/F')$, trivial on $\mathrm{Gal}(\overline{F}'/F')^e$, corresponds to one, V'', of $\mathrm{Gal}(\overline{F}''/F'')$, and if Ψ' corresponds to Ψ'', we have*

$$\varepsilon(V', \Psi', dx) = \varepsilon(V'', \Psi'', dx)\ . \tag{3.7.1.1}$$

It concerns the local constant ε of the functional equation of L-functions, as defined in [2].

Suppose first V' and V'' are of dimension 1 corresponding to characters χ' and χ'' of F'^* and F''^*. By hypothesis, they factor respectively through $F'^*/U_{F'}^e$ and $F''^*/U_{F''}^e$. By 3.6.1, χ' and χ'' correspond via the isomorphism induced by u

$$F'^*/U_{F'}^e \simeq (\mathrm{Tr}_e F')^* \simeq (\mathrm{Tr}_e F'')^* \simeq F''^*/U_{F''}^e\ :$$

In arbitrary dimension, the characters of F'^* and F''^* associated to the maximal exterior powers $\det V'$ and $\det V''$ of V' and V'' correspond similarly. From there, and from the formulas

$$\varepsilon(V, \Psi, a \cdot dx) = a^{\dim V} \varepsilon(V, \Psi, dx)$$

$$\varepsilon(V, \Psi(\lambda x), dx) = \det V(\lambda) \|\lambda\|^{-\dim V} \varepsilon(V, \Psi, dx)$$

we deduce that the validity of (3.7.1.1) does not depend on the choice of dx, nor on that of Ψ', Ψ'' which correspond.

Let us return to dimension 1. If χ', hence χ'', is unramified, if dx gives volume 1 to the valuation ring and if $n(\Psi') = n(\Psi'') = 0$, we have $\varepsilon = 1$ and (3.7.1.1) is true. Suppose χ' ramified, of conductor $e(1) \leq e$. Let us choose Ψ' such that $n(\Psi') = -e(1)$ and dx giving volume 1 to $\mathfrak{m}'^{e(1)}$. The constant $\varepsilon(\chi', \Psi', dx)$ is then the sum over $(\mathcal{O}'/\mathfrak{m}'^{e(1)})^*$

$$\sum \chi(x)^{-1}\Psi(x) \ ,$$

and the same sum gives $\varepsilon(\chi'', \Psi'', dx)$. This proves (3.7.1.1) in dimension 1.

Let $\overline{F}'$ be a separable closure of F', $\overline{F}'_e$ the subfield of $\overline{F}'$ fixed by $\mathrm{Gal}(\overline{F}'/F')^e$, $E' \subset \overline{F}'$ an extension of F' in $(\mathrm{ext}\ F')^e$, i.e. contained in $\overline{F}'_e$, $e(1) = \Psi_{E'/F'}(e)$ and W' a virtual representation of dimension 0 of $\mathrm{Gal}(\overline{F}'_e/E') = \mathrm{Gal}(\overline{F}'/E')/\mathrm{Gal}(\overline{F}'/E')^{e(1)} = \mathrm{Gal}(\overline{F}'/E')/\mathrm{Gal}(\overline{F}'/E') \cap \mathrm{Gal}(\overline{F}'/F')^e$, and V' the induced virtual representation of $\mathrm{Gal}(\overline{F}'_e/F')$. Let E'', W'' and V'' corresponding to E', W' and V'. One has an isomorphism $u_1 : \mathrm{Tr}_{e(1)}E' \simeq \mathrm{Tr}_{e(1)}E''$ and W' corresponds to W'' rel. this isomorphism.

Lemma 0.17. *3.7.2. If additive characters Ψ' and Ψ'' of F' and F'' correspond (rel. u), the characters $\Psi' \circ Tr_{E'/F'}$ and $\Psi'' \circ Tr_{E''/F''}$ correspond (rel. u_1).*

Set $(R, M, \varepsilon) := \mathrm{Tr}_e(F')$ and $(A_1, I_1, \varepsilon) = (\mathrm{Tr}_{re}(E'))_1$. We can assume that $n(\Psi') = n(\Psi'') = -(e+1)$. In this case, Ψ' matters to us only by its restriction to $M = \mathfrak{m}_{F'}/\mathfrak{m}_{F'}^{e+1}$. We have $n(\Psi' \circ \mathrm{Tr}_{E'/F'}) = rn(\Psi') - (\text{valuation of the different}) = -(\Psi_{E'/F'}(e) + 1)$: the character $\Psi' \circ \mathrm{Tr}_{E'/F'}$ matters to us only by its restriction to $I_1 = \mathfrak{m}_{E'}/\mathfrak{m}_{E'}^{e(1)+1}$. This is given by

$$(\Psi' \mid M) \circ (\text{trace } 3.4.4).$$

Similarly for Ψ'', and 3.7.2 follows.

We have

$$\varepsilon(V', \Psi') = \varepsilon(W', \Psi' \circ \mathrm{Tr}_{E'/F'})$$

(these ε are independent of dx), and similarly with $'$ replaced by $''$. From $\varepsilon(W', \Psi' \circ \mathrm{Tr}_{E'/F'}) = \varepsilon(W'', \Psi'' \circ \mathrm{Tr}_{E''/F''})$, we can thus conclude that $\varepsilon(V', \Psi') = \varepsilon(V'', \Psi'')$. According to 3.7.2, and the dimension 1 case, already treated, such is the case if W' is an algebraic sum of representations of dimension 1. We conclude by using that, in the Grothendieck group $\bar{R}(G)$ of virtual representations of a finite group G, every element x is a linear combination

$$x = n \cdot 1 + \sum n_i \mathrm{Ind}_{H_i}^G(\chi_i - 1),$$

with χ_i a representation of dimension 1 of $H_i \subset G$.

Remark 2 (3.7.3). The comparison statement 3.7.1 also holds for the L factors.

3.8. If F' and F'' have algebraically closed residue fields, geometric class field theory [7] provides an isomorphism

$$(\mathrm{Gal}(\overline{F}'/F')/\mathrm{Gal}(\overline{F}'/F')^e)^{\mathrm{ab}} \simeq \pi_1(\mathcal{O}_{F'}^*/U^e(F')) \quad .$$

An isomorphism u as in 3.4. provides an isomorphism of pro-algebraic groups of $\mathcal{O}_{F'}^*/U^e(F')$ with $\mathcal{O}_{F''}^*/U^e(F'')$. We have

Proposition 0.18 (3.8.1). <u>*The diagram*</u>

$$\begin{array}{ccc} (Gal(\overline{F}'/F')/Gal(\overline{F}'/F')^e)^{ab} & \xrightarrow[\sim]{U} & (Gal(\overline{F}''/F'')/Gal(\overline{F}''/F'')^e)^{ab} \\ \downarrow & & \downarrow \\ \pi_1(\mathcal{O}_{F'}^*/U^e(F')) & \xrightarrow[\sim]{u} & \pi_1(\mathcal{O}_{F''}^*/U^e(F'')) \end{array}$$

<u>*is commutative*</u>

One proceeds as for 3.6.1, by using the description of the reciprocity isomorphism [7] remark p. 119.

APPENDIX. RAMIFICATION THEORY AND HERBRAND FUNCTIONS, FOR NON-GALOIS EXTENSIONS

The theory of Herbrand functions φ and Ψ, as presented by J.P. Serre in [6] IV, extends without difficulty to the case of separable, not necessarily Galois, extensions E/F. For lack of finding a convenient reference in the literature, we explain this extension here.

(A.1) Let F be a complete field with respect to a discrete valuation, with perfect residue field, and E a finite separable extension of F. If E is Galois over F, with Galois group G, J.P. Serre defines the ramification groups G_n by

(A.1.1) and Herbrand's Theorem

$$\begin{aligned} \sigma \in G_n : &\iff \forall x \in \mathcal{O}_E \ (\sigma(x) \equiv x \mod \mathfrak{m}_E^{n+1}) \\ &\iff \forall x \in \mathfrak{m}_E - \mathfrak{m}_E^2 \ (\sigma(x)/x \equiv 1 \mod \mathfrak{m}_E^n) \quad . \end{aligned} \tag{A.1.1}$$

This is the lower numbering. Note carefully that in this numbering the inertia group is G_0, whereas it is G_1 in that of Zariski-Samuel vol I chapter V §10. For real $u \geq -1$, one sets

$$G_u := G_n \quad \text{for } n \text{ the smallest integer } \geq u,$$

and defines the Herbrand function $\varphi_{E/F}$ or simply φ by

$$\varphi(u) = \int_0^u dt/[G_0 : G_t] \quad . \tag{A.1.2}$$

We adopt the convention that $\varphi(u) = u$ for $-1 \leq u \leq 0$. For $H \subset G$ a normal subgroup and $E' = E^H$ the corresponding Galois sub-extension of E, with Galois group G/H, one proves the transitivity

$$\varphi_{E/F} = \varphi_{E'/F} \circ \varphi_{E/E'} \tag{A.1.3}$$

and Herbrand's theorem

$$(G/H)_v = (G_u H)/H \quad \text{for } v = \varphi_{E/E'}(u) \quad . \tag{A.1.4}$$

(A.2) Remarks on Notation

(A.2) Before giving the generalization of these facts to extensions which are not necessarily Galois, two remarks which will simplify the notation.

1. For E' and E'' two extensions of F, with ramification indices r' and r'', we will denote by $v_{E'}$ the multiple $\frac{r'}{r''} v_{E''}$ of the valuation $v_{E''}$ of E'' normalized so that its value group is $\mathbb{Z}$. For any embedding $\iota : E'' \longrightarrow E'''$, we have $v_{E'}(\iota x) = v_{E'}(x)$.

2. Let us recall the remark of Serre [6] §3 p. 83, elaborated in [1] 3.1. To each finite separable extension E of F, with group of orders $\Gamma_E := E^*/\mathcal{O}_E^* \xrightarrow[\sim]{v_E} \mathbb{Z}$, let us attach the half-line $T_E : t \geq -1$ of $\Gamma_E \otimes \mathbb{R} \xrightarrow{\sim} \mathbb{R}$. For E' a Galois extension of E'', we view the Herbrand function $\varphi_{E'/E''}$ as an isomorphism $T_{E'} \xrightarrow{\sim} T_{E''}$. One deduces from (A.1.2) that these isomorphisms generate a transitive system of isomorphisms between the T_E. Let T be the projective limit of the T_E. We denote by v_E the composite map $T \xrightarrow{\sim} T_E \hookrightarrow \Gamma_E \otimes \mathbb{R} \xrightarrow[\sim]{v_E} \mathbb{R}$.

(A.3) Let E be a finite separable extension of F. Let M be a finite Galois extension of F, large enough for E to embed into it. The Galois group used in (A.1) is to be replaced by the set G of embeddings of E into M. It is a homogeneous space under $\mathrm{Gal}(M/F)$. The filtration by the G_u is to be replaced by the data of equivalence relations $R(t)$ on G ($t \in T$), increasingly fine as $t \to \infty$. Denoting by $v_E(\sigma - \tau)$ the infimum of the $v_E(\sigma(x) - \tau(x))$ for $x \in \mathcal{O}_E$, we define them by

$$\sigma \equiv \tau \pmod{R(t)} :\iff v_E(\sigma - \tau) \geq v_E(t) + 1 \quad . \tag{A.3.1}$$

Note that G, equipped with the $R(t)$, does not change when M is replaced by $M' \supset M$. This remark would allow replacing the choice of M by that of a separable closure of $\bar{F}$. One defines upper and lower numberings by

$$R(t) = R^{v_F(t)} = R_{v_E(t)} \quad . \tag{A.3.2}$$

In lower numbering, one simply has

$$\sigma \equiv \tau \pmod{R_u} \iff \forall x \in \mathcal{O}_E \;\; v_E(\sigma(x) - \tau(x)) \geq u + 1 \quad . \tag{A.3.3}$$

The Galois group $\mathrm{Gal}(M/F)$ acting transitively on G and respecting the equivalences $R(t)$, the number $r(t)$ of elements in a class mod $R(t)$ is independent of the chosen class. We set

$$r(t) = r^{v_F(t)} = r_{v_E(t)} \quad . \tag{A.3.4}$$

Let x be a generator of $\mathcal{O}_E$ over $\mathcal{O}_F$. According to [6] III 6 prop.12, such a generator exists. If $P(X) = X^d + \sum a_i X^i$ is its characteristic polynomial – it coincides with its minimal polynomial – the homomorphism $X \longmapsto x : \mathcal{O}_F[X]/(P(X)) \longrightarrow \mathcal{O}_E$ is an isomorphism and the map $\sigma \mapsto \sigma(x)$ thus identifies G with the set of roots of P in M. For $\sigma, \tau \in G$, we have

$$v_E(\sigma - \tau) = v_E(\sigma(x) - \tau(x)) \quad . \tag{A.3.5}$$

The integers r_t can then be read from the valuations of the roots of the polynomial $P(X + x)$, with coefficients in $\mathcal{O}_E$: once an embedding σ_o of E into M is chosen, the roots of this polynomial in the extension M of E are the $\sigma(x) - \sigma_o(x)$ ($\sigma \in G$). The valuations of the roots of $P(X + x)$ can

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can be read from its Newton polygon.

(A.4) Let us fix an embedding σ_o of E into M. Remark (A.2)b leads to defining the Herbrand function $\varphi_{E/F}$ as the composite

$$\varphi_{E/F} := \varphi_{M/F} \circ \varphi_{M/E}^{-1} \tag{A.4.1}$$

Proposition 0.19. ***A.4.2.*** *The function (A.4.1) is given by the formula, parallel to (A.1.2)*

$$\varphi(u) = \int_0^u dt/(r_o : r_t) \tag{A.4.3}$$

Proof. It suffices to show that the functions (A.4.3) obey the transitivity (A.1.3). The essential point is the following generalization of [6] IV §1 Prop. 3 p. 71. Let $E \subset E'$, and M be large enough for E' to embed in it. The restriction to E is then a surjection from $G' := \mathrm{Hom}_F(E', M)$ onto $G := \mathrm{Hom}_F(E, M)$. □

Proposition 0.20. ***A.4.4.*** *For $\sigma_o, \sigma_1 \in G$, for τ_o a lifting of σ_o and for τ_1 running through the liftings of σ_1, we have*

$$v_E(\sigma_o - \sigma_1) = \sum_{\tau_1 \to \sigma_1} v_E(\tau_o - \tau_1)$$

The proof is parallel to that of [6]. We choose generators x and y of $\mathcal{O}_E$ and $\mathcal{O}_{E'}$ over $\mathcal{O}_F$. If Q is the minimal polynomial of y over E, we have

$$\sigma Q = \prod_{\tau \to \sigma} (X - \tau y)$$

Taking $\sigma = \sigma_1$, and substituting $X = \tau_o y$, we find

$$\prod_{\tau_1 \to \sigma_1} (\tau_o y - \tau_1 y) = (\sigma_1 Q)(\tau_o y) \equiv \sigma_o Q(\tau_o y) = 0 \pmod{\sigma_1 x - \sigma_o x}$$

In the other direction, we write $x = G(y)$, for $G \in \mathcal{O}_F[X]$. Since y is a root of $x - G(X)$, we have $Q \mid x - G(X) : x - G(X) = Q(X) \cdot h(X)$. Applying σ_1, and substituting $X = \tau_o y$, we find

$$\sigma_1 x - \sigma_o x = (\sigma_1 Q)(\tau_o y) \cdot (\sigma_1 h)(\tau_o y) \equiv 0 \pmod{(\sigma_1 Q)(\tau_o y)} = \prod_{\tau_1 \to \sigma_1} (\tau_o y - \tau_1 y).$$

These two divisibilities prove the assertion.

This being established, the transitivity of the functions (A.4.3) is proved as in [6] IV §3 Proposition 15 page 81. One proves at the same time ([6] IV §3 Lemma 5 page 82) that for $E \subset E'$ as in (A.4), the equivalence relation $R(t)$ on G is the quotient of the equivalence relation $R(t)$ on G'. In particular, taking $E' = M$, one finds that $R(t)$ is the equivalence: "being in the same orbit of $\mathrm{Gal}(M/F)(t) = \mathrm{Gal}(M/F)_{v_M(t)} = \mathrm{Gal}(M/F)^{v_F(t)}$".

A.5. Let E/F, M and $G := \mathrm{Hom}_F(E, M)$ be as in (A.3). Let ℓ be the largest real number such that the equivalence R_ℓ is non-trivial. It is the supremum of the $v_E(\sigma - \tau) - 1$ for $\sigma \neq \tau$ in G. For $u \geq \ell$, the function $\varphi_{E/F}(u)$ is linear, with slope the inverse of the ramification index r of E/F. By analogy with (A.2)a., let us set for $\sigma, \tau \in G$

$$v_F(\sigma - \tau) = \frac{1}{r} v_E(\sigma - \tau) = \inf\{v_F(\sigma(x) - \tau(x)) \mid x \in \mathcal{O}_E\} \quad .$$

Proposition 0.21. *Fix $\sigma_0 \in G$. We have*

$$\varphi_{E/F}(\ell) + 1 = \sum_{\sigma \neq \sigma_0} v_F(\sigma - \sigma_0) + \sup_{\sigma \neq \sigma_0} v_F(\sigma - \sigma_0) \quad . \tag{A.5.2}$$

Proof. In the sum $\sum_{\sigma \neq \sigma_0} v_E(\sigma - \sigma_0)$, let us group the equal terms and write the obtained sum over t as a Stieltjes integral extended from -1 to ∞ or from -1 to $\ell + 0$, which amounts to the same thing:

$$\sum_{\sigma \neq \sigma_0} v_E(\sigma - \sigma_0) = \sum_t (t+1)(r_t - r_{t+0}) = \int_{-1}^{\ell+0} -(t+1) dr_t \quad .$$

Let us integrate by parts:

$$\begin{aligned} &= \int_{-1}^{\ell} dt.r_t - [(t+1).r_t]_{-1}^{\ell+0} \\ &= r_o \int_{-1}^{\ell} dt/(r_o : r_t) - (\ell + 1) \\ &= r_o(\varphi(\ell) + 1) - (\ell + 1) \quad . \end{aligned}$$

Dividing by the ramification index $r = r_o$, we obtain (A.5.2). □

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For $u \geq \ell$, we have $\varphi_{E/F}(u) + 1 = \varphi_{E/F}(\ell) + 1 + \frac{1}{r}(u - \ell) = [(\varphi_{E/F}(\ell) + 1) - \frac{1}{r}(\ell + 1)] + \frac{1}{r}(u + 1)$, that is by (A.5.2)

$$\text{(A.5.3)} \quad \varphi_{E/F}(u) + 1 = \frac{1}{r}(u+1) + \frac{1}{r} \sum_{\sigma \neq \sigma_0} v_F(\sigma - \sigma_0) \qquad (u \geq \ell)\,,$$

and for the inverse function $\Psi_{E/F}$

$$\text{(A.5.4)} \quad \Psi_{E/F}(f) + 1 = r(f+1) - \sum_{\sigma \neq \sigma_0} v_E(\sigma - \sigma_0) \qquad (f \geq \varphi_{E/F}(\ell))\,.$$

The second term on the right-hand side is the valuation of the different. If f is an integer $\geq \varphi_{E/F}(\ell)$, $\Psi_{E/F}(f)$ is therefore an integer.

Proposition 0.22. *A.6.1. For every real number f, the following conditions are equivalent:*
(i) The equivalence relation R^f is trivial.
(ii) The normal closure E' of E satisfies $Gal(E'/E)^f = \{e\}$.
(iii) For ℓ as in A.5, $\varphi_{E/F}(\ell) < f$.
(iv) $\sum_{\sigma \neq \sigma_0} v_F(\sigma - \sigma_0) + \sup_{\sigma \neq \sigma_0} v_F(\sigma - \sigma_0) < f + 1$.

To prove (i) $\iff$ (ii), we take $M = E'$ and use that the classes mod R^f are the orbits of $\mathrm{Gal}(M/F)^f$ (A.4). That (i) $\iff$ (iii) is the definition of ℓ and (iii) $\iff$ (iv) by A.5.1.

Let P be the characteristic polynomial of a generator x of O_E over O_F (cf. (A.3)). The following proposition reinterprets (A.6.1) (iv) in terms of the approximate roots of P.

Proposition 0.23. *A.6.2. For the equivalent conditions of (A.6.1) to be satisfied, it is necessary and sufficient that for every y in a separable closure $\bar{F}$ of F satisfying $v_{\bar{F}}(P(y)) \geq f+1$, there exists a root z of P closer than the others: $v_{\bar{F}}(y-z) > v_{\bar{F}}(y-z')$ for z' another root of P.*

Proof. If z is a root of P as close to y as possible ($v(y-z)$ maximal), we have for every root z' □

$$v_{\bar{F}}(y-z') = \inf(v_{\bar{F}}(y-z), v_{\bar{F}}(z'-z)),$$

hence

$$v_{\bar{F}}P(y) = v_{\bar{F}}(y-z) + \sum_{z' \neq z} \inf(v_{\bar{F}}(y-z), v_{\bar{F}}(z'-z)).$$

Comparing with (A.5.2) by using (A.3.5), one finds that for the root z to be closer to y than the others, it is necessary and sufficient that

$$v_{\bar{F}}P(y) > \varphi_{E/F}(\ell) + 1.$$

That this is always a consequence of $v_{\bar{F}}P(y) \geq f+1$ is equivalent to $\varphi_{E/F}(\ell) < f$.

Remark 3 (A.6.3). Let ℓ be as in A.5 : $\ell + 1 = \sup v_E(\sigma - \sigma_o)$ and $L = \varphi_{E/F}(\ell)$. Let z be a root of P in $\bar{F}$. According to A.6.2, the map $y \longmapsto P(y)$ induces a bijection from the disk $\{y \mid v_E(y-z) > \ell+1\}$ to the disk $\{a \mid v_F(a) > L+1\}$. For each approximate solution y : $v_{\bar{F}}P(y) > L+1$, the root z of P closer to y than the others can be obtained by Newton's method, and $v_{\bar{F}}(y-z) = v_F(P(y)/P'(z))$. In terms of the function $\Psi_{E/F}$ inverse of $\varphi_{E/F}$; we have by (A.5.4) : if f satisfies the equivalent conditions of A.6.1 and A.6.2, the map $y \longmapsto P(y)$ induces a bijection

$$\{y \mid v_E(y-z) \geq \Psi_{E/F}(f)+1\} \xrightarrow{\sim} \{a \mid v_F(a) \geq f+1\}.$$

Remark 4 (A.6.4). With the same hypothesis on f, if y is an approximate solution of $P = 0$ in an extension E of F, in the sense $v_F P(y) \geq f+1$, the root y_o of P closest to y, being obtained by Newton's method, will still be in F, and $v_E(y-y_o) \geq \Psi_{E/F}(f)+1$. Conversely, if $v_E(y-y_o) \geq \Psi_{E/F}(f)+1$, then $v_F P(y) \geq f+1$.

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